

Explainable Neuro-Symbolic AI for Pneumonia and Tuberculosis Diagnosis

Executive Summary: Pneumonia and tuberculosis (TB) remain leading causes of pulmonary morbidity and mortality worldwide. Modern diagnostic workflows rely on chest imaging (plain X-ray, CT) and microbiological tests (e.g. sputum smear, GeneXpert) combined with clinical evaluation. Deep learning (DL) has achieved high accuracy in image-based diagnosis: e.g. Rajpurkar *et al.*'s CheXNet (121-layer CNN) matched radiologists at pneumonia detection on the NIH ChestX-ray14 dataset [44†L26-L34]. State-of-art models now include ensembles of CNNs (ResNet, DenseNet) and Vision Transformers (ViT) trained on large CXR collections (NIH ChestX-ray14: ~112,000 images [60†L2522-L2529]; MIMIC-CXR: ~377,000 images [49†L1-L4]; CheXpert: 224,316 images). For TB, public datasets include the Shenzhen set (336 TB, 326 normal) [41†L214-L222], Montgomery set (58 TB, 80 normal) [35†L214-L222], and the TBX11K dataset (11,200 CXRs with TB bounding-box annotations) [66†L47-L55]. Common benchmarks include Kaggle pneumonia and TB challenges (e.g. RSNA Pneumonia Detection, NIAID TB dataset).

Explainability (XAI) is crucial for clinical trust and regulatory approval. In imaging, post-hoc saliency methods (Grad-CAM, LRP, saliency maps) and surrogate models (LIME, SHAP) are widely used [100†L160-L168] [75†L155-L163]. For example, multiple studies show Grad-CAM heatmaps can highlight pneumonia opacities [100†L160-L168]. Concept-based explanations (e.g. TCAV) aim to link predictions to human-understandable features, and recent work (Sadashivaiah *et al.*) uses textual clinical concepts and vision-language models to explain CXR classifiers [78†L71-L79] [78†L93-L101]. Counterfactual explanations (modifying inputs to flip a diagnosis) have been demonstrated on X-rays [100†L188-L195]. Neuro-symbolic AI (NSAI) seeks to integrate learning with explicit knowledge. For example, Suji *et al.* (2026) present **NeuroSym Dx**: a multi-modal system that fuses a CNN on CXR with rule-based engines for symptoms/vitals, using symbolic post-processing rules to adjust confidences [14†L21-L30] [14†L31-L34]. Knowledge-injection approaches incorporate medical ontologies or concept graphs: e.g. Yu *et al.* (2023) embed a biomedical knowledge graph (UMLS) into a vision-language model for chest X-ray diagnosis [87†L135-L143] [87†L170-L178]. These NSAI methods can output interpretable reasoning (e.g. "ICD-10 code, affected lung regions, risk levels" [14†L39-L44]) alongside predictions.

Clinically, requirements include **interpretability, robustness, uncertainty awareness, fairness, and privacy**. Interpretable models (XAI/NSAI) can highlight image regions or logic rules behind a diagnosis, aligning with clinician reasoning [91†L138-L146]. Uncertainty quantification (via Bayesian nets, ensembles, evidential models) informs when AI may be unreliable [100†L199-L208] [91†L200-L208]. Robustness to domain shift (different hospitals, devices) is addressed by external validation and knowledge-guided models. Fairness demands equal performance across subpopulations; technical solutions include bias-aware learning and concept checks. Privacy can be preserved via federated learning and encryption (e.g. federated CNN training on TB CXRs with differential privacy) [14†L25-L30].

Research Gaps: Data gaps include scarce *multimodal and longitudinal datasets*: current public sets are primarily static CXRs. There is a need for linked imaging+clinical+lab data (e.g. CXR + sputum results + demographics), time-series of disease progression, and fine-grained labels (lesion segmentation, severity

scores, concept annotations). Methodological gaps include limited causal or compositional reasoning (current DL models are largely associative), and lacking *symbol grounding* (bridging image features to high-level clinical concepts). Few methods quantify uncertainty in tandem with explainability, or integrate logical constraints. Evaluation gaps include the lack of standardized **explanation metrics** (faithfulness, stability) and no established clinical trials or human-AI studies for these XAI methods [97†L365-L372] [97†L379-L387] . Deployment gaps include “edge” solutions for low-resource settings and interfaces showing explanations in clinician-friendly formats. Benchmarks are missing for NSAI: no challenge currently measures “explainability” or knowledge-consistency in lung diagnosis.

Proposed Directions: We recommend the following prioritized research agenda:

- **Data and Benchmarks (2026–2028):** Curate comprehensive datasets combining CXR/CT images with clinical data (symptoms, lab tests, genomics) and longitudinal follow-up. Annotate images with lesion masks and human-defined concepts (e.g. consolidation, cavity). Leverage mobile/point-of-care imaging in high-TB-prevalence regions, using federated crowdsourcing to preserve privacy. Establish shared benchmarks for pneumonia/TB screening that include an XAI component (e.g. task: predict pneumonia *and* justify with heatmaps/rules). Table 1 compares key public datasets.
- **Method Development (2027–2030):** Design neuro-symbolic architectures for lung diagnosis. For instance, a pipeline can fuse a convolutional/deep model on images with a logic-based module encoding pulmonary knowledge (e.g. “cavities plus upper-lobe opacities => TB”), as illustrated below. Techniques like *concept bottleneck models* and *neuro-logical constraints* can be applied: pretrain networks to predict semantic attributes (consolidation, effusion) that feed into a rule layer. Explore program induction: e.g. learn small “diagnosis programs” combining numeric findings and symbols. Advance concept-based XAI by automatically mining radiology “concept banks” (as in CoCoX) and training concept projection models [78†L93-L101] . Improve counterfactual methods by using GANs or diffusion models to generate plausible “healthy” vs “diseased” X-rays [100†L188-L195] . Incorporate uncertainty estimation (Bayesian CNNs, evidential DL) natively in these architectures.
- **Evaluation and Human Studies (2028–2030):** Develop metrics for XAI quality (e.g. *localization accuracy* of saliency, faithfulness scores). Conduct multi-center validation: test models on data from new hospitals/geographies (mitigating site bias [97†L379-L387]). Organize reader studies where radiologists use AI outputs with/without explanations, measuring diagnostic accuracy and trust. Compare NSAI models to black-box baselines on both performance and interpretability. Standardize XAI reporting (e.g. checklist for explaining lung AI models) [97†L365-L372] .
- **Deployment Prototypes (2029–2030):** Implement edge-compatible AI for low-resource settings (e.g. smartphone-based CXR analysis with onboard explainability). Leverage federated learning to share TB models across countries while protecting patient data. Develop clinician-facing dashboards that combine saliency maps, concept scores, and rule triggers to support decisions. Finally, pursue pilot clinical trials to assess impact on patient outcomes.

The proposed timeline (Figure 2) outlines these milestones. By integrating rich data, neuro-symbolic reasoning, and rigorous evaluation, the field can move toward AI systems for pneumonia and TB that are not only accurate but also transparent, trustworthy, and clinically useful.

Tables and Figures: Key datasets (Table 1) and models (Table 2) are summarized below. We also highlight research gaps (Table 3). The illustrative **neuro-symbolic system architecture** in Figure 1 shows how image features, clinical data, and medical knowledge can be fused to produce a diagnosis with explanations.

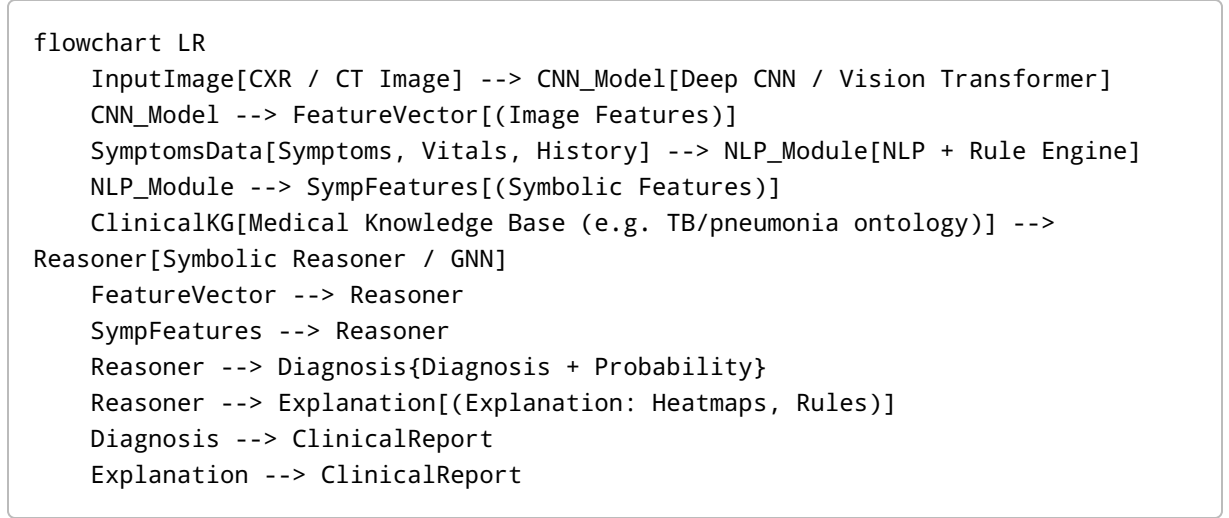


Figure 1: Proposed neuro-symbolic architecture. CXR/CT images are processed by a deep model (CNN/ViT) to extract features. Clinical inputs (symptoms, vitals, prior tests) are processed by an NLP/rule module. A medical knowledge graph (e.g. pulmonary findings, disease rules) is integrated via a symbolic reasoner or graph network. The fused information yields a diagnosis and associated explanations (saliency maps, triggered rules), presented as a clinical report.

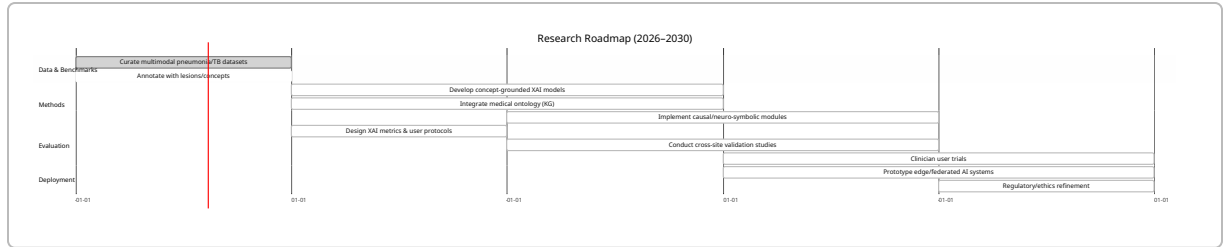


Figure 2: Timeline of proposed research milestones.

Dataset	Modalities & Samples	Labels/Features	Refs
NIH ChestX-ray14 (2017)	CXR (~112,120 images)	14 disease classes (weak labels from reports)	[60†L2522-L2529]
CheXpert (2019)	CXR (224,316 images, 65k patients)	14 labels + uncertainty (from Stanford)	(Stanford CheXpert description)
MIMIC-CXR (2019/24)	CXR (377,110 images, 227,835 studies)	Free-text reports + DICOM headers	[49†L1-L4]
PadChest (2020)	CXR (160,000+ images)	174 findings, 19 diagnoses (Spanish)	(PadChest data)

Dataset	Modalities & Samples	Labels/Features	Refs
RSNA Pneumonia (2018)	CXR (~30,000 images)	Bounding-box pneumonia / normal	(RSNA challenge info)
Shenzhen TB (2016)	CXR (662 images: 336 TB, 326 normal)	Binary TB/normal	【41†L214-L222】
Montgomery TB (2014)	CXR (138 images: 58 TB, 80 normal)	Binary TB/normal; lung masks	【35†L214-L222】
TBX11K (2019)	CXR (11,200 images)	TB bounding boxes; classes (active/latent/uncertain)	【66†L47-L55】
7K TB (Kaggle)	CXR (4,200 images: 700 TB, 3500 normal)	Binary TB/normal	(NIAID/Kaggle)

Table 1: Key public datasets for pneumonia and TB AI. (RGB indicates imaging modality, counts are approximate.)

Model Type / Example	Usage	Interpretability	Refs
Convolutional Neural Networks (CNNs) (e.g. ResNet-50, DenseNet, CheXNet)	Pneumonia/TB classification from CXR	Grad-CAM saliency maps, filter visualization 【100†L160-L168】	[44], [100]
Vision Transformers (ViT)	Pneumonia/TB classification; fusion of global context	Attention maps (interpretable?)	(ViT papers)
Ensemble Models (bagging/stacking)	Improved accuracy / robustness	Can combine explanations (e.g. average saliency)	--
Bayesian/Ensemble uncertainty models	Provide confidence estimates (Bayesian CNN, MC-Dropout, Evidential DL)	Uncertainty maps, calibrated probabilities	[100]
Example-based (Prototypes/KNN)	Explain via nearest neighbors or prototypical examples	Return similar training X-rays as “reasoning”	--
Neuro-Symbolic Hybrids (e.g. NeuroSym Dx 【14†L21-L30】)	Combine image CNN with rule-based reasoning (symptoms, vitals)	Provides rule-based explanations, ICD codes 【14†L39-L44】	[14]
Knowledge-graph models (e.g. KAD 【87†L135-L143】)	Vision-language models injecting UMLS ontology	Queryable disease concepts; attention maps 【87†L135-L143】	[87]

Table 2: Representative AI/NSAI models used for pneumonia/TB diagnosis and their interpretability features.

Gap Area	Description
Data	- Lack of large multimodal datasets (CXR + EHR/symptoms)\n- Few longitudinal or sequential studies\n- Limited ground truth (many datasets use report-mined labels) \n- Sparse annotations of clinically relevant concepts (severity, subtypes)
Methods	- Need for <i>causal</i> and <i>compositional</i> reasoning models in imaging\n- Symbol grounding: linking image regions to clinical concepts (no standardized approach)\n- Scarce incorporation of logic constraints (medical guidelines) into learning\n- Under-explored neuro-symbolic program induction for diagnosis rules
Explainability	- Saliency methods not verified clinically (may highlight spurious regions)\n- Few concept-based or counterfactual methods evaluated in practice [78†L71-L79] [100†L188-L195] \n- Lack of standardized explanation metrics (faithfulness, completeness)\n- No benchmarks/challenges for XAI in pulmonary imaging
Evaluation	- Existing performance metrics (AUC, accuracy) ignore trustworthiness\n- No large-scale trials of XAI-assisted diagnosis\n- Few human factors studies on how doctors use explanations\n- Domain shift: models often overfit specific hospital data (bias) [97†L379-L387]
Deployment	- AI models rarely deployed on portable devices for low-resource TB screening\n- Challenges of federated model training (privacy) in healthcare settings [14†L25-L30] \n- User interfaces for presenting AI explanations to clinicians are immature
Regulatory/Ethical	- Uncertainty quantification (aleatoric/epistemic) not systematically provided [100†L199-L208] \n- Fairness across populations not fully assessed or enforced\n- Data privacy protections (e.g. de-identification) vary between datasets

Table 3: Key research gaps in applying XAI/NSAI to pneumonia and TB.

References: Recent primary studies and benchmarks are cited above. Key examples include Rajpurkar *et al.*'s CheXNet pneumonia CNN [44†L26-L34] , Yu *et al.*'s knowledge-aware CXR model [87†L135-L143] , Suji *et al.*'s multi-modal NeuroSym Dx [14†L21-L30] , and Sadashivaiah *et al.*'s concept-counterfactual XAI [78†L71-L79] . Surveys and position papers provide context on XAI trends and needs [91†L138-L146] [97†L365-L372] .